

Data Communication and Networks

Lecture 1

Introduction and Overview

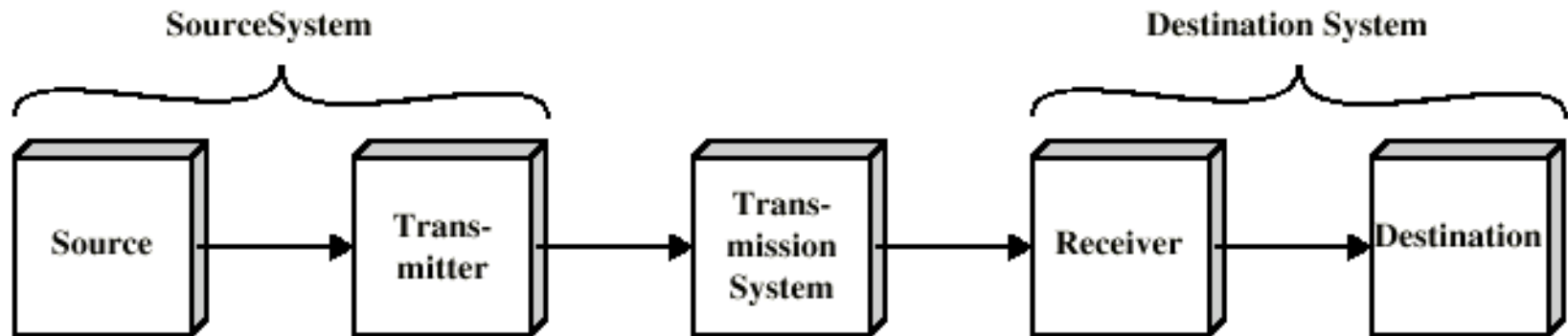
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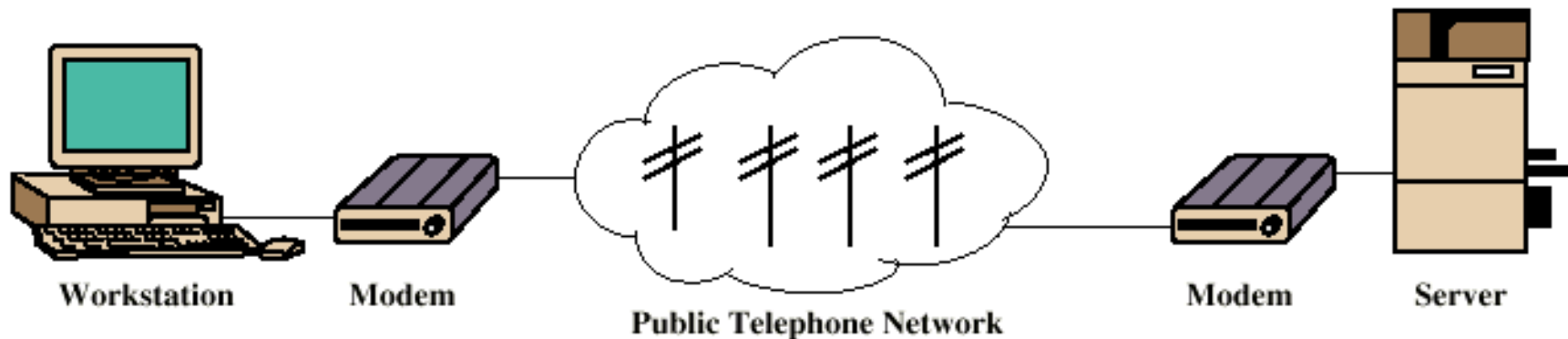
A Communications Model

- Source
 - generates data to be transmitted
- Transmitter
 - Converts data into transmittable signals
- Transmission System
 - Carries data
- Receiver
 - Converts received signal into data
- Destination
 - Takes incoming data

Simplified Communications Model - Diagram



(a) General block diagram

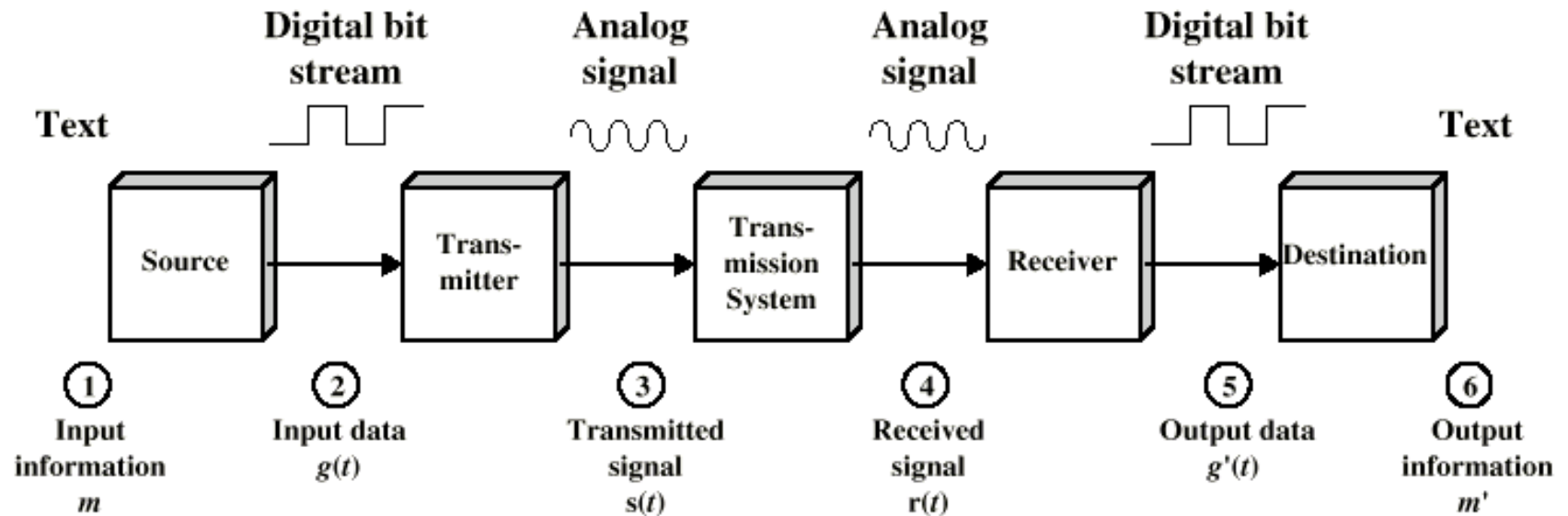


(b) Example

Key Communications Tasks

- Transmission System Utilization
- Interfacing
- Signal Generation
- Synchronization
- Exchange Management
- Error detection and correction
- Addressing and routing
- Recovery
- Message formatting
- Security
- Network Management

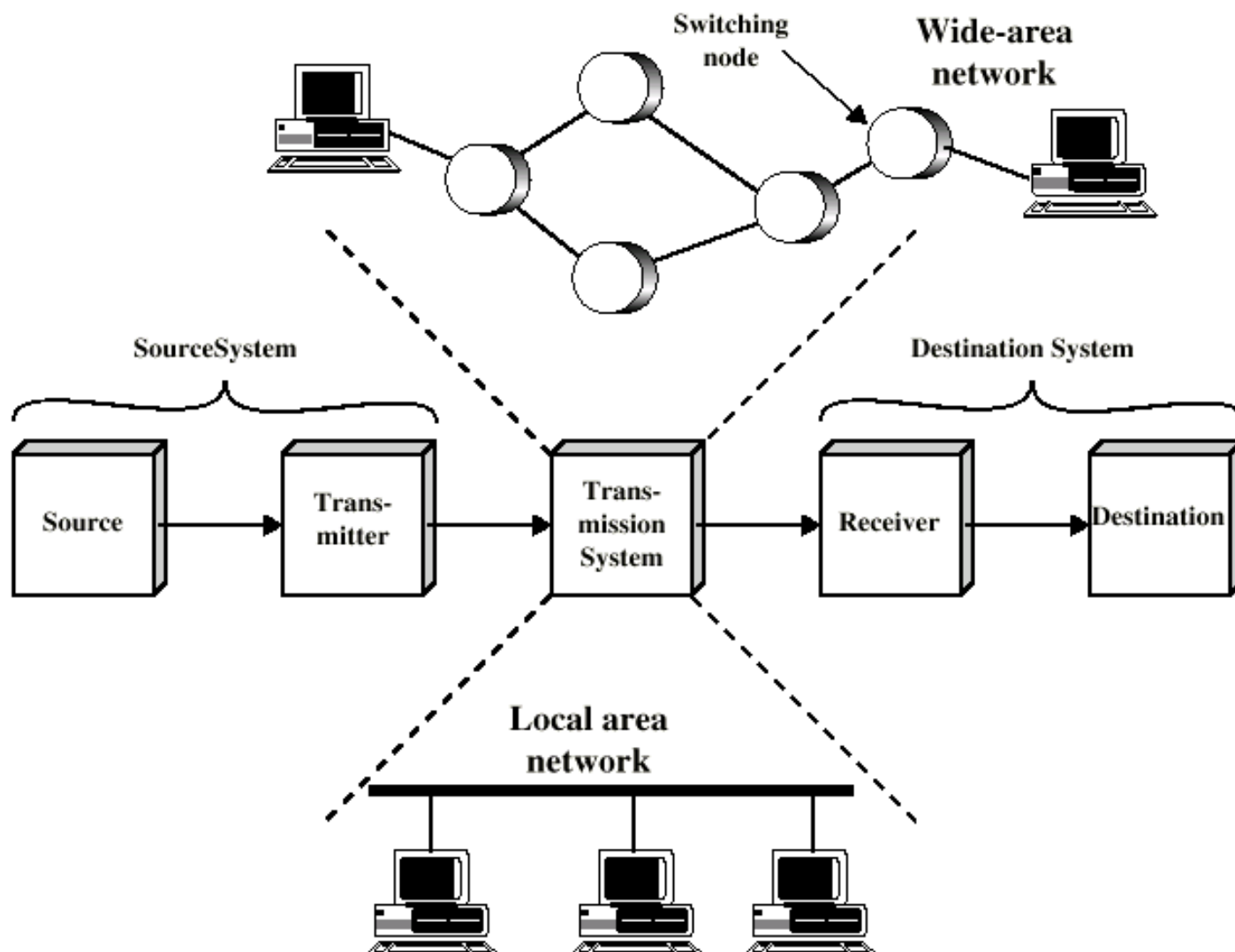
Simplified Data Communications Model



Networking

- Point to point communication not usually practical
 - Devices are too far apart
 - Large set of devices would need impractical number of connections
- Solution is a communications network

Simplified Network Model



Wide Area Networks

- ❑ Large geographical area
- ❑ Crossing public rights of way
- ❑ Rely in part on common carrier circuits
- ❑ Alternative technologies
 - ❑ Circuit switching
 - ❑ Packet switching
 - ❑ Frame relay
 - ❑ Asynchronous Transfer Mode (ATM)

Circuit Switching

- Dedicated communications path established for the duration of the conversation
- e.g. telephone network

Packet Switching

- Data sent out of sequence
- Small chunks (packets) of data at a time
- Packets passed from node to node between source and destination
- Used for terminal to computer and computer to computer communications

Frame Relay

- ❑ Packet switching systems have large overheads to compensate for errors
- ❑ Modern systems are more reliable
- ❑ Errors can be caught in end system
- ❑ Most overhead for error control is stripped out

Asynchronous Transfer Mode

- ATM
- Evolution of frame relay
- Little overhead for error control
- Fixed packet (called cell) length
- Anything from 10Mbps to Gbps
- Constant data rate using packet switching technique

Local Area Networks

- Smaller scope
 - Building or small campus
- Usually owned by same organization as attached devices
- Data rates much higher
- Usually broadcast systems
- Now some switched systems and ATM are being introduced

Protocols

- Used for communications between entities in a system
- Must speak the same language
- Entities
 - User applications
 - e-mail facilities
 - terminals
- Systems
 - Computer
 - Terminal
 - Remote sensor

Key Elements of a Protocol

- Syntax
 - Data formats
 - Signal levels
- Semantics
 - Control information
 - Error handling
- Timing
 - Speed matching
 - Sequencing

What's a protocol?

human protocols:

- “what’s the time?”
- “I have a question”
- introductions

... specific msgs sent

... specific actions taken when
msgs received, or other
events

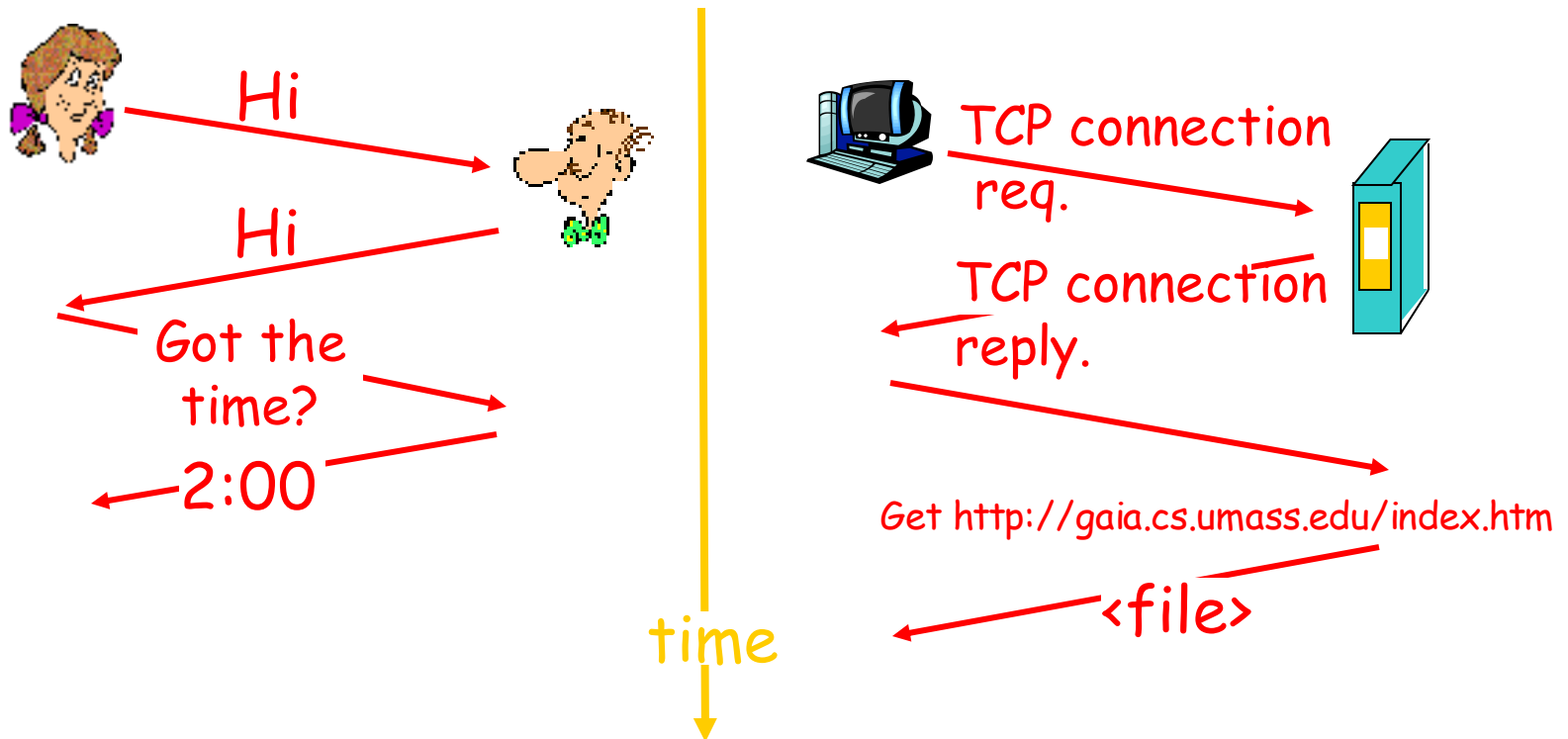
network protocols:

- machines rather than humans
- all communication activity in
Internet governed by protocols

*protocols define format, order of msgs
sent and received among network
entities, and actions taken on msg
transmission, receipt*

What's a protocol?

a human protocol and a computer network protocol:



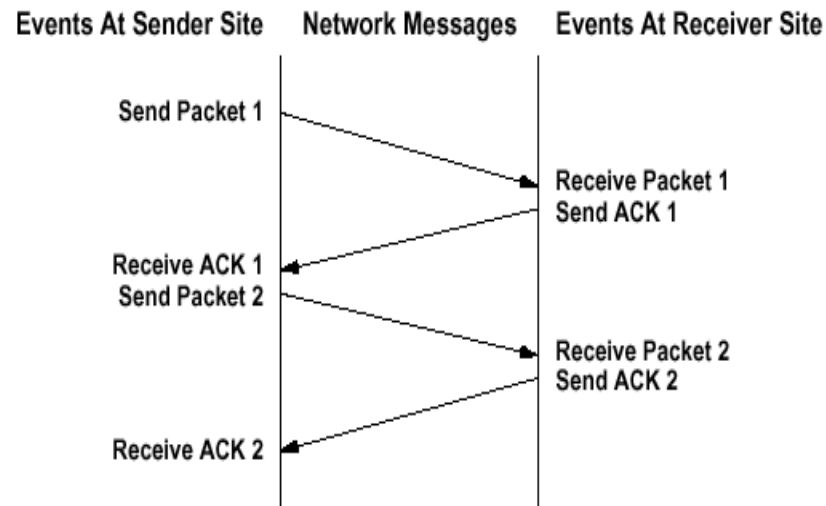
In Summary, a protocol is

- An agreement about communication between two or more entities
- It specifies
 - Format of messages
 - Meaning of messages
 - Rules for exchange
 - Procedures for handling problems

Protocol Specification

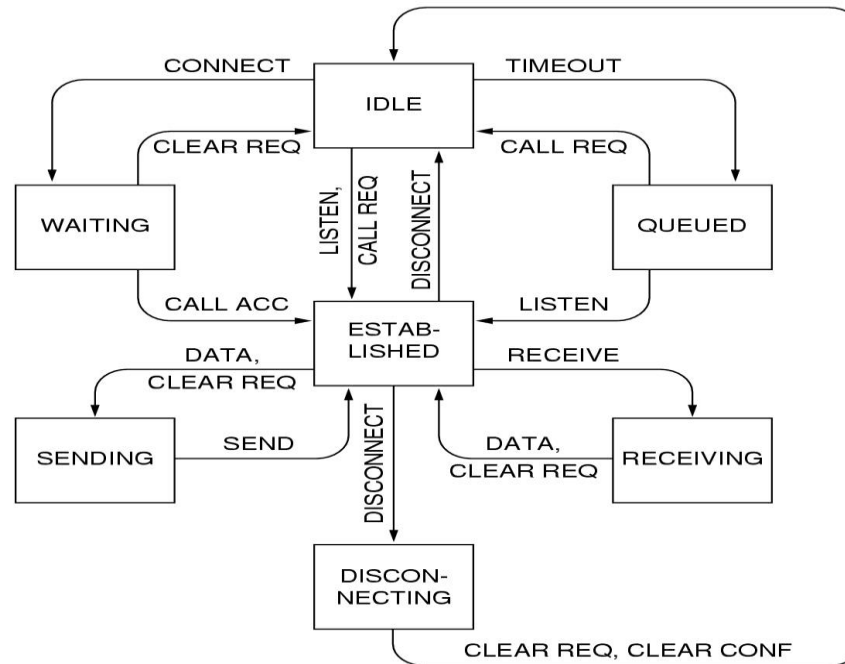
- As designers, we can specify a protocol using
 - Space-Time Diagrams
 - Transition Diagrams
- We can implement a protocol with a Finite State Machine (FSM)
- Internet Protocols are formalized by RFCs which are administered by IETF

Space-Time Diagrams



- Defines causal ordering
- Defines indication/request/response actions

Transition Diagram

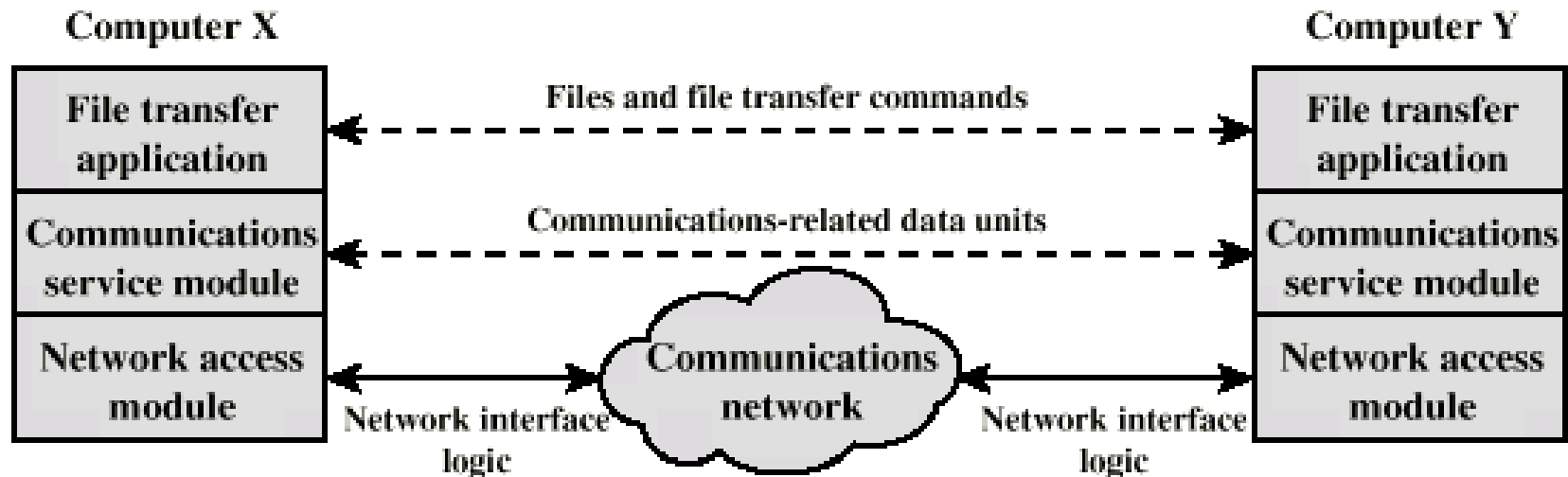


- **Illustrates**
 - States
 - Input (the event that causes transition)
 - Transitions (to new states)

Protocol Architecture

- Task of communication broken up into modules
- For example file transfer could use three modules
 - File transfer application
 - Communication service module
 - Network access module

Simplified File Transfer Architecture



A Three Layer Model

- Network Access Layer
- Transport Layer
- Application Layer

Network Access Layer

- Exchange of data between the computer and the network
- Sending computer provides address of destination
- May invoke levels of service
- Dependent on type of network used (LAN, packet switched etc.)

Transport Layer

- Reliable data exchange
- Independent of network being used
- Independent of application

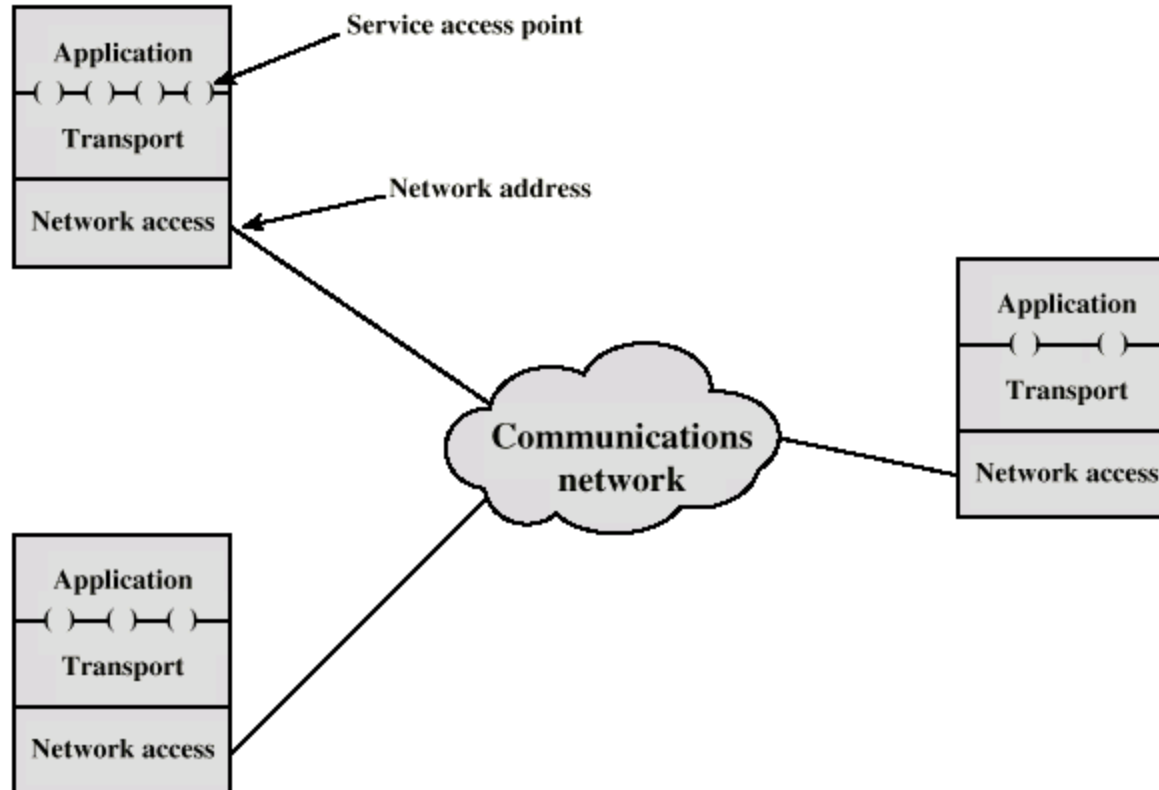
Application Layer

- Support for different user applications
- e.g. e-mail, file transfer

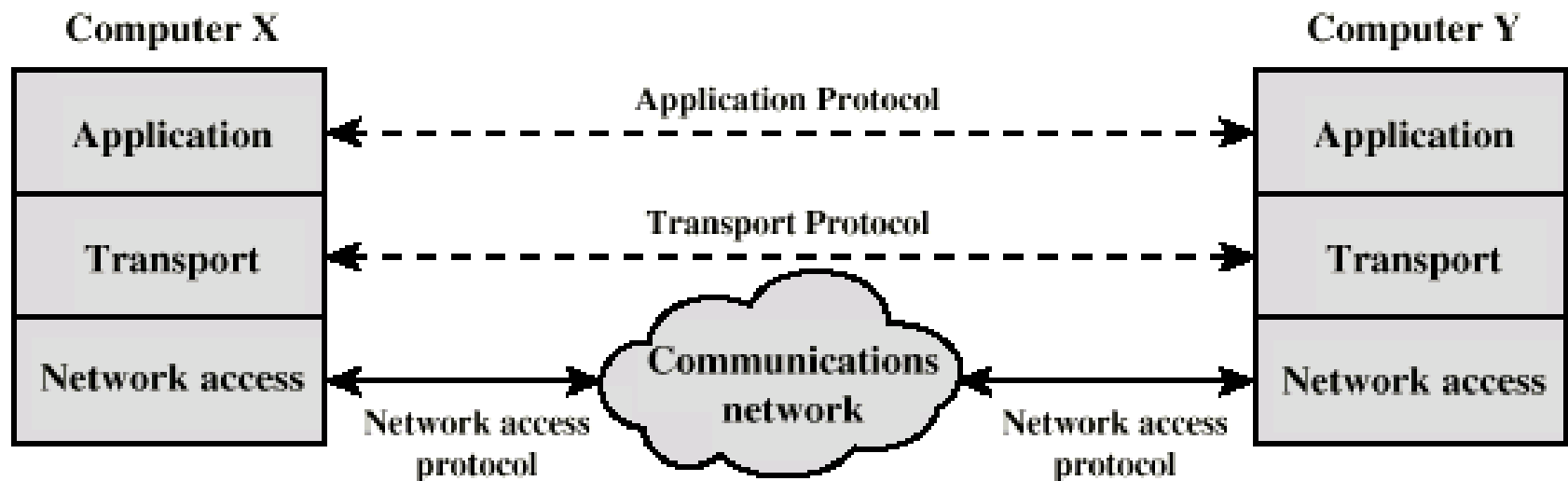
Addressing Requirements

- Two levels of addressing required
- Each computer needs unique network address
- Each application on a (multi-tasking) computer needs a unique address within the computer
 - The service access point or SAP
 - The port on TCP/IP stacks

Protocol Architectures and Networks



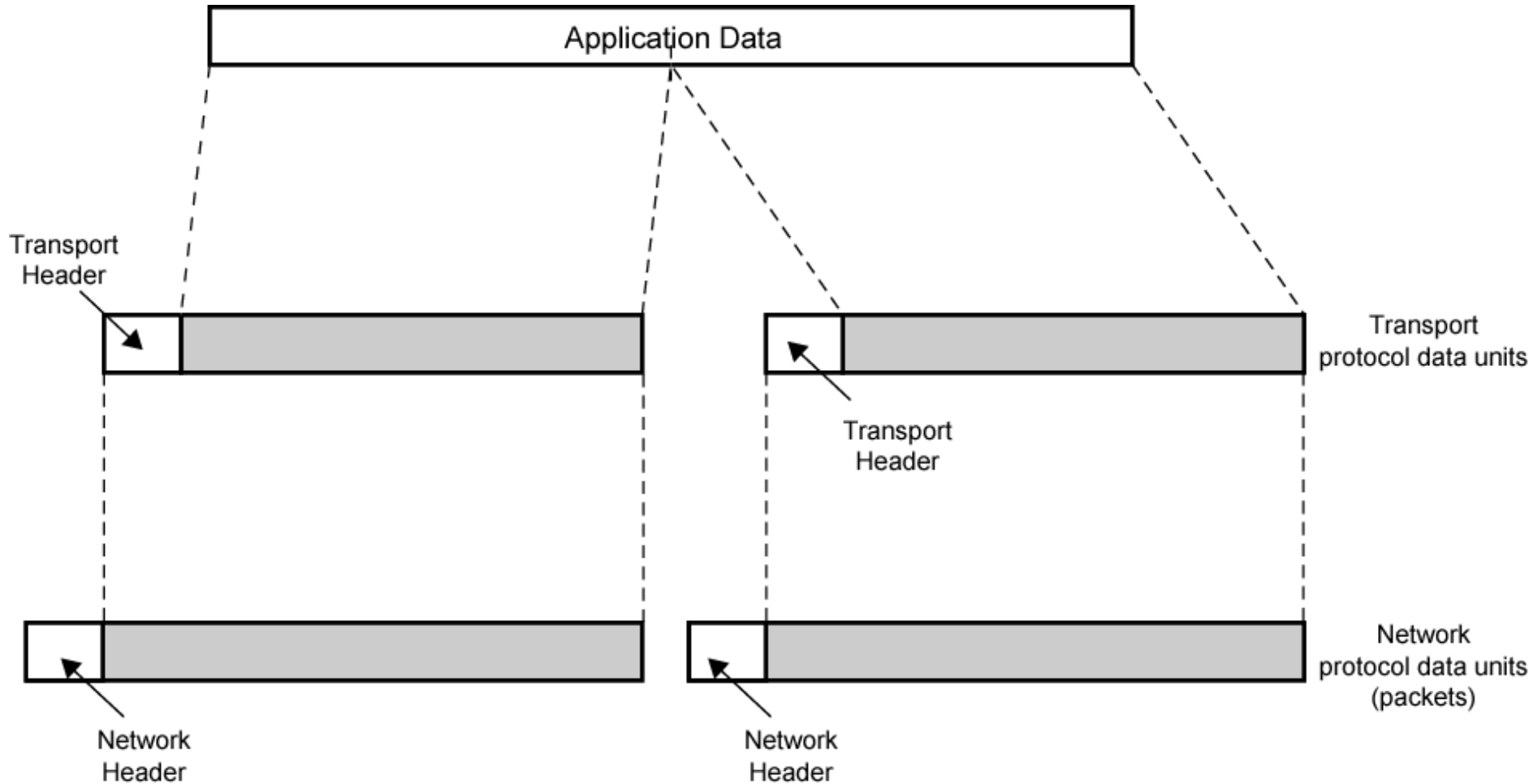
Protocols in Simplified Architecture



Protocol Data Units (PDU)

- At each layer, protocols are used to communicate
- Control information is added to user data at each layer
- Transport layer may fragment user data
- Each fragment has a transport header added
 - Destination SAP
 - Sequence number
 - Error detection code
- This gives a transport protocol data unit

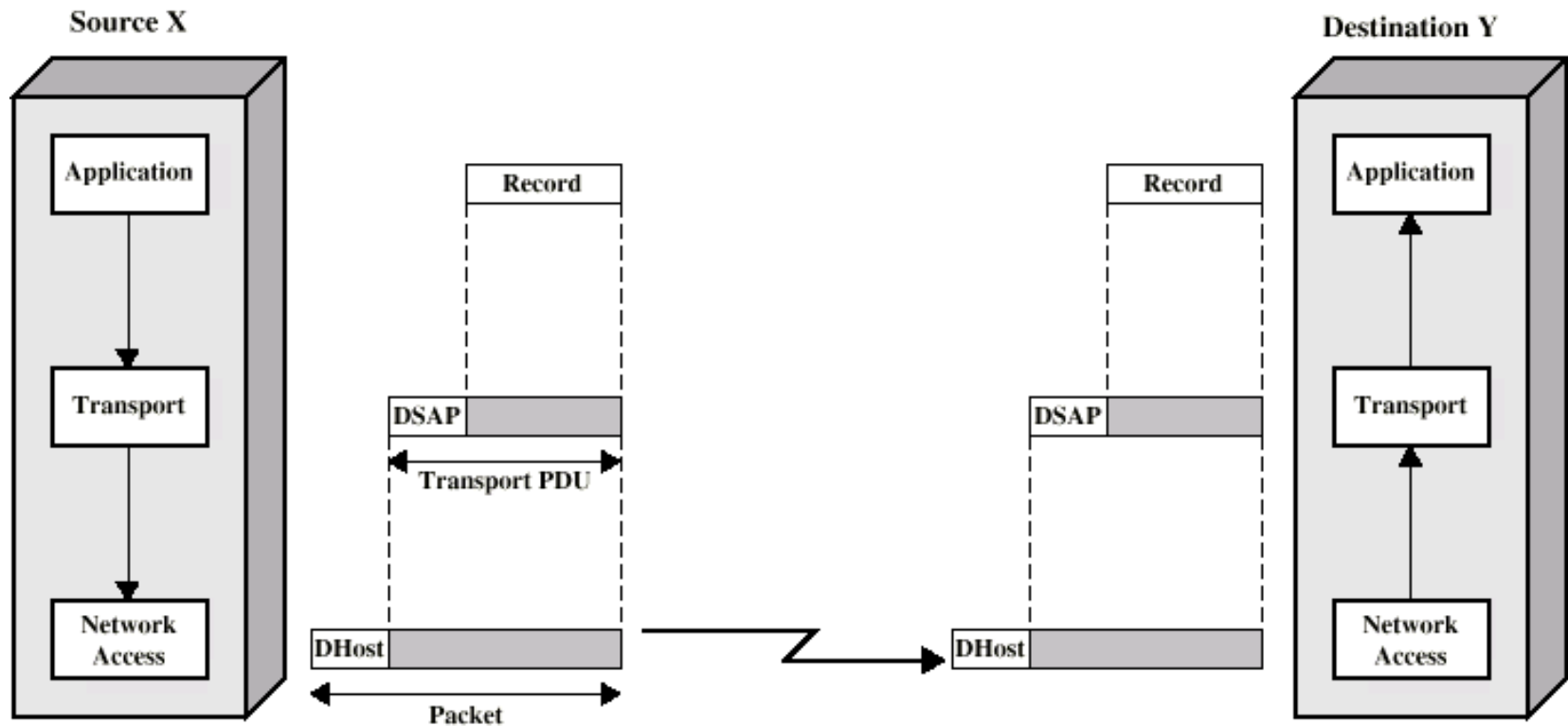
Protocol Data Units



Network PDU

- Adds network header
 - network address for destination computer
 - Facilities requests

Operation of a Protocol Architecture



Standardized Protocol Architectures

- ❑ Required for devices to communicate
- ❑ Vendors have more marketable products
- ❑ Customers can insist on standards based equipment
- ❑ Two standards:
 - ❑ OSI Reference model
 - ❑ Never lived up to early promises
 - ❑ TCP/IP protocol suite
 - ❑ Most widely used
- ❑ Also: IBM Systems Network Architecture (SNA)

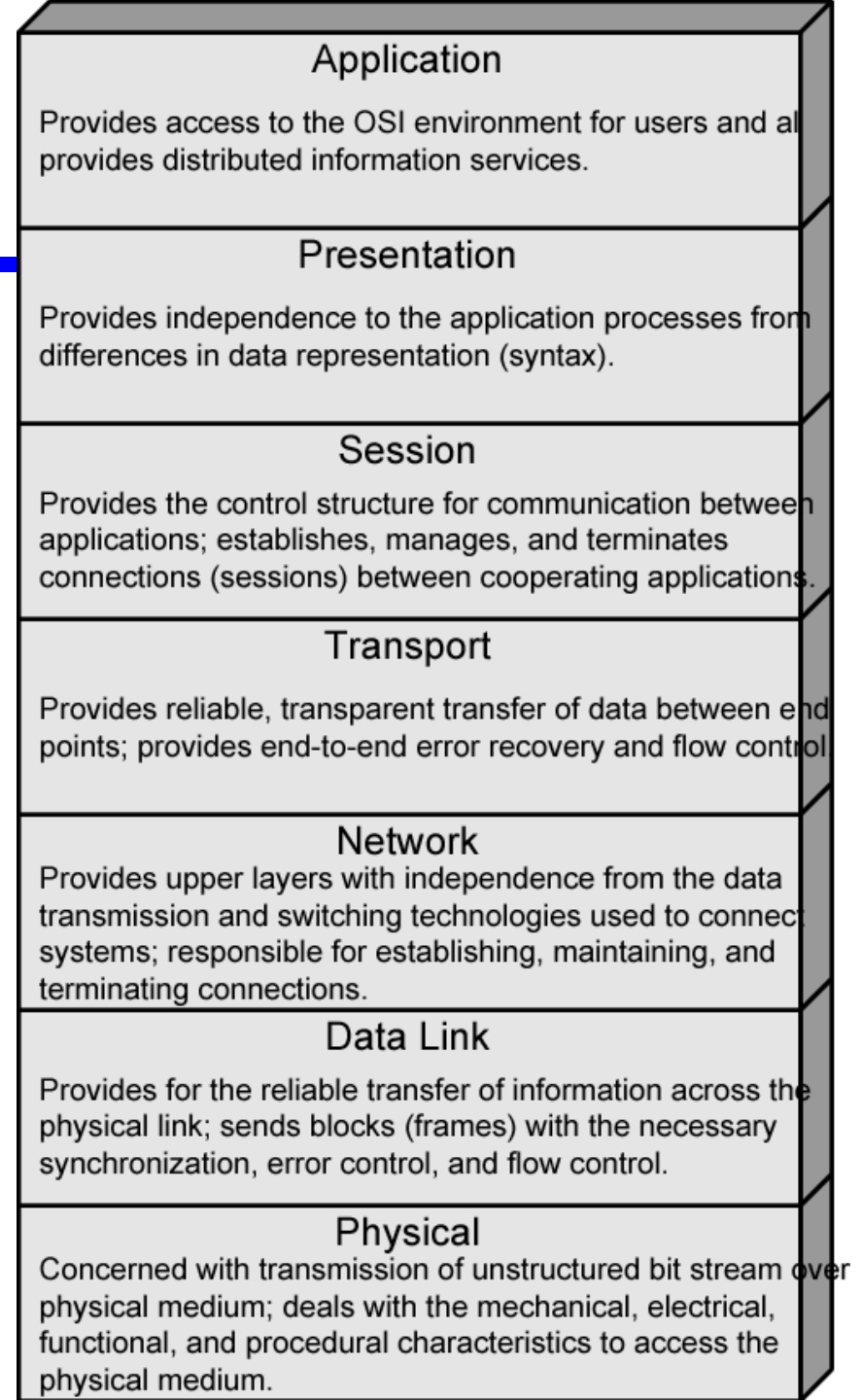
OSI

- ❑ Open Systems Interconnection
- ❑ Developed by the International Organization for Standardization (ISO)
- ❑ Seven layers
- ❑ A theoretical system delivered too late!
- ❑ TCP/IP is the de facto standard

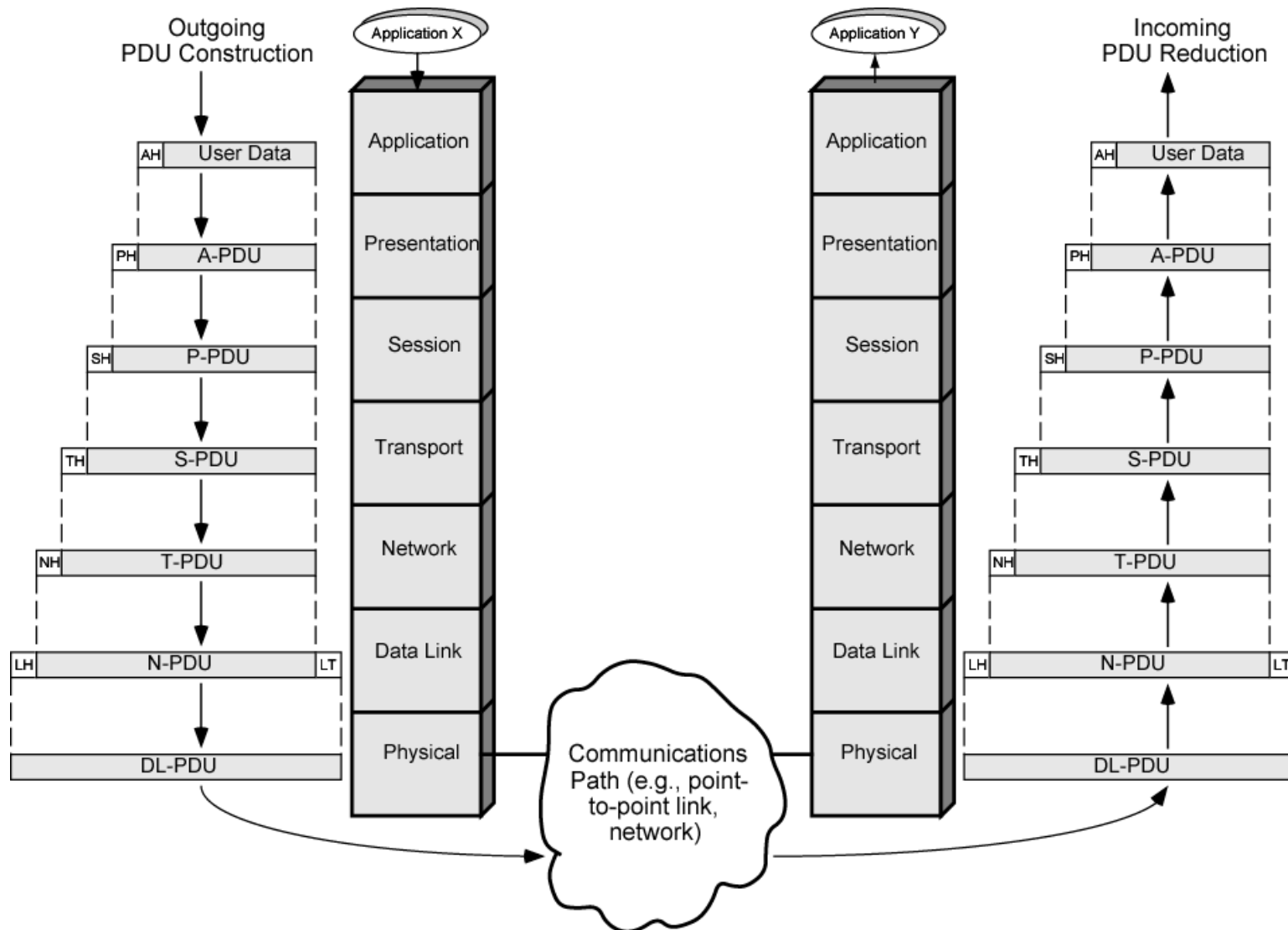
OSI - The Model

- A layer model
- Each layer performs a subset of the required communication functions
- Each layer relies on the next lower layer to perform more primitive functions
- Each layer provides services to the next higher layer
- Changes in one layer should not require changes in other layers

OSI Layers



The OSI Environment



TCP/IP Protocol Architecture

- ❑ Developed by the US Defense Advanced Research Project Agency (DARPA) for its packet switched network (ARPANET)
- ❑ Used by the global Internet
- ❑ No official model but a working one.
 - ❑ Application layer
 - ❑ Host to host or transport layer
 - ❑ Internet layer
 - ❑ Network access layer
 - ❑ Physical layer

Physical Layer

- Physical interface between data transmission device (e.g. computer) and transmission medium or network
- Characteristics of transmission medium
- Signal levels
- Data rates
- etc.

Network Access Layer

- Exchange of data between end system and network
- Destination address provision
- Invoking services like priority

Internet Layer (IP)

- Systems may be attached to different networks
- Routing functions across multiple networks
- Implemented in end systems and routers

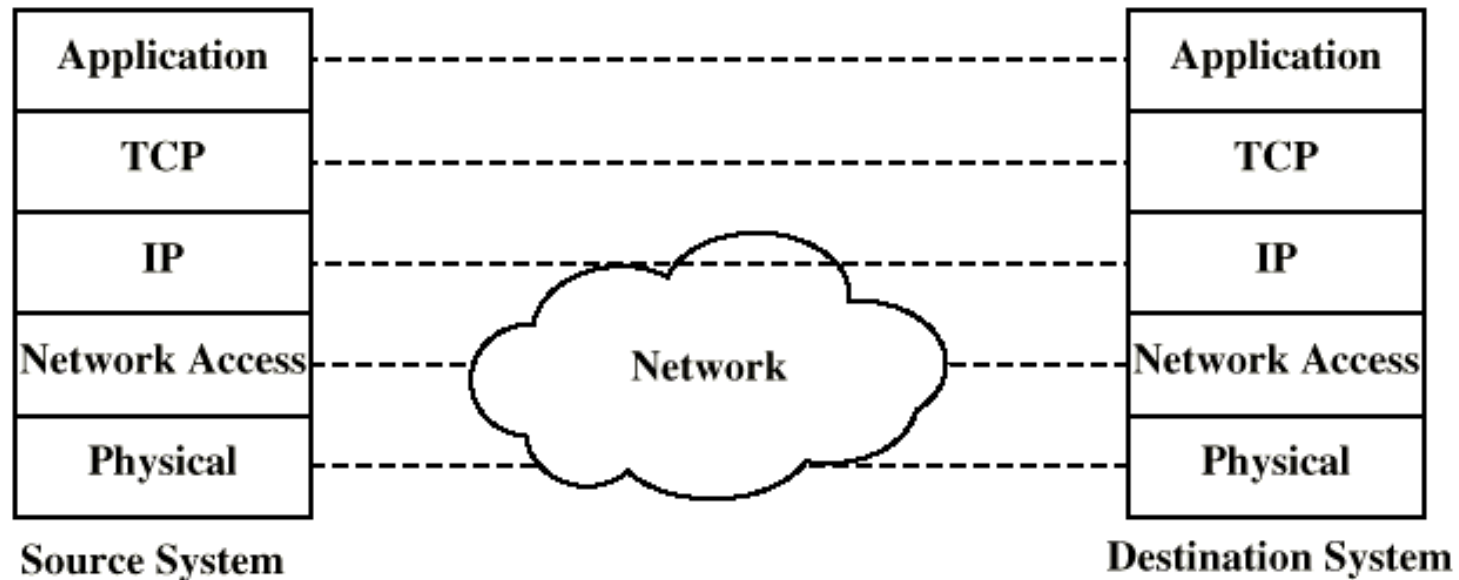
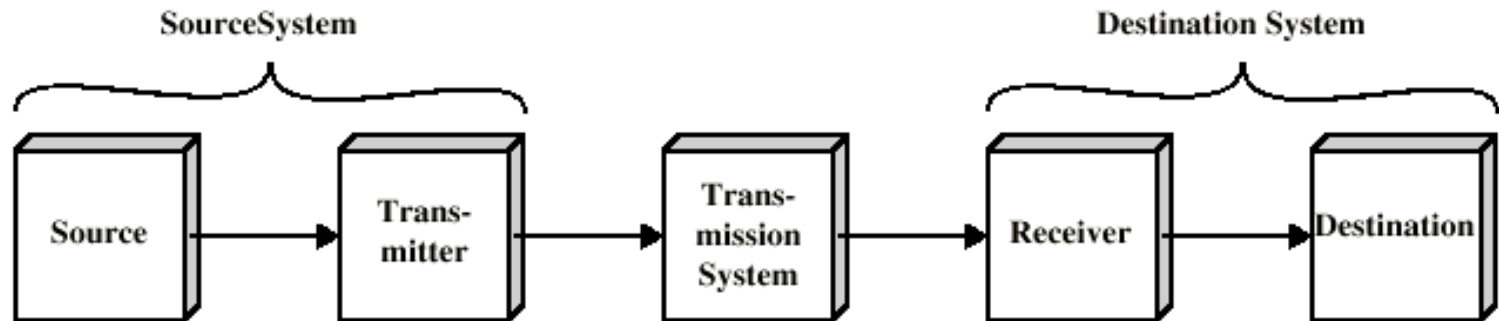
Transport Layer (TCP)

- Reliable delivery of data
- Ordering of delivery

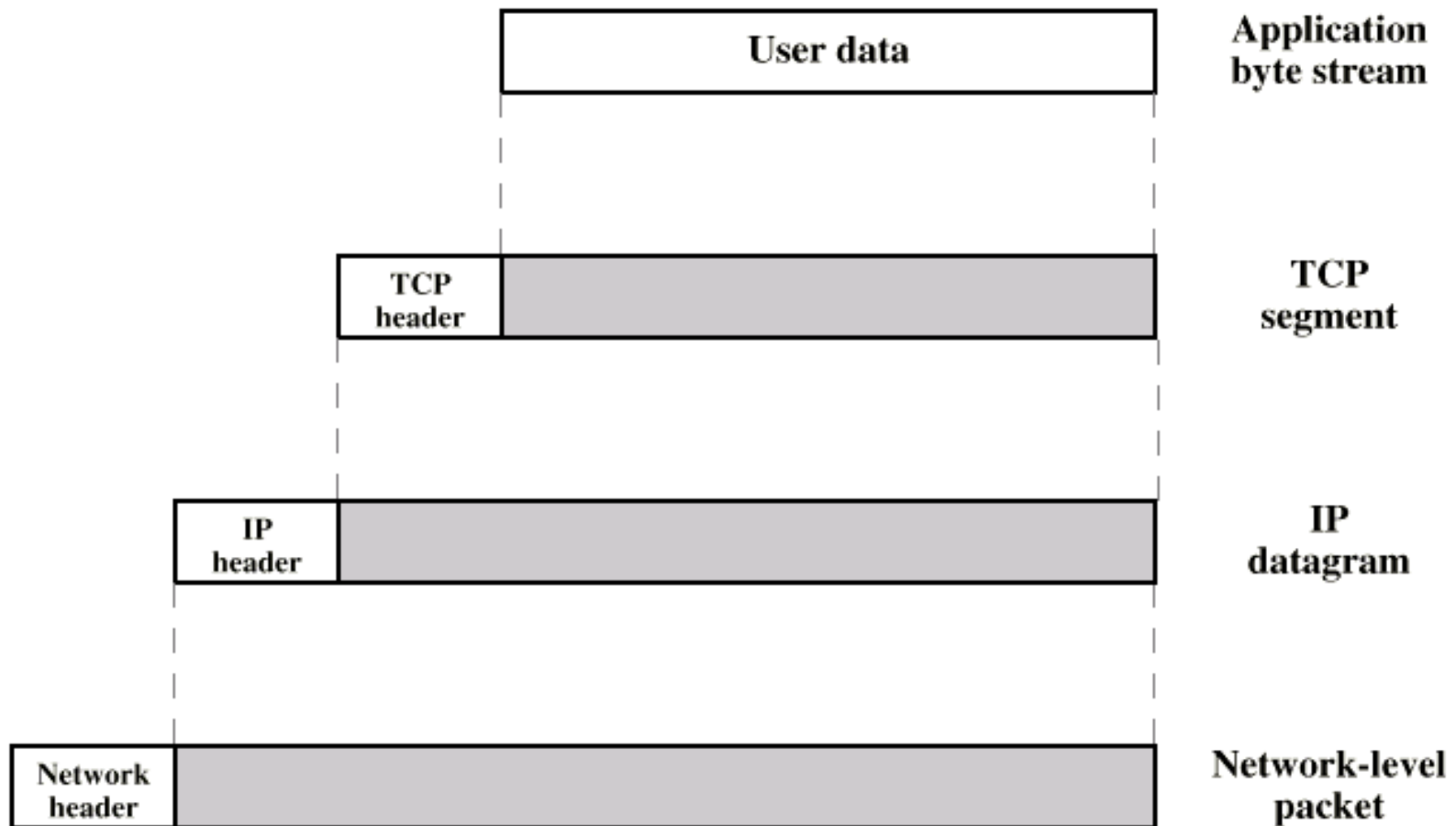
Application Layer

- Support for user applications
- e.g. http, SMTP

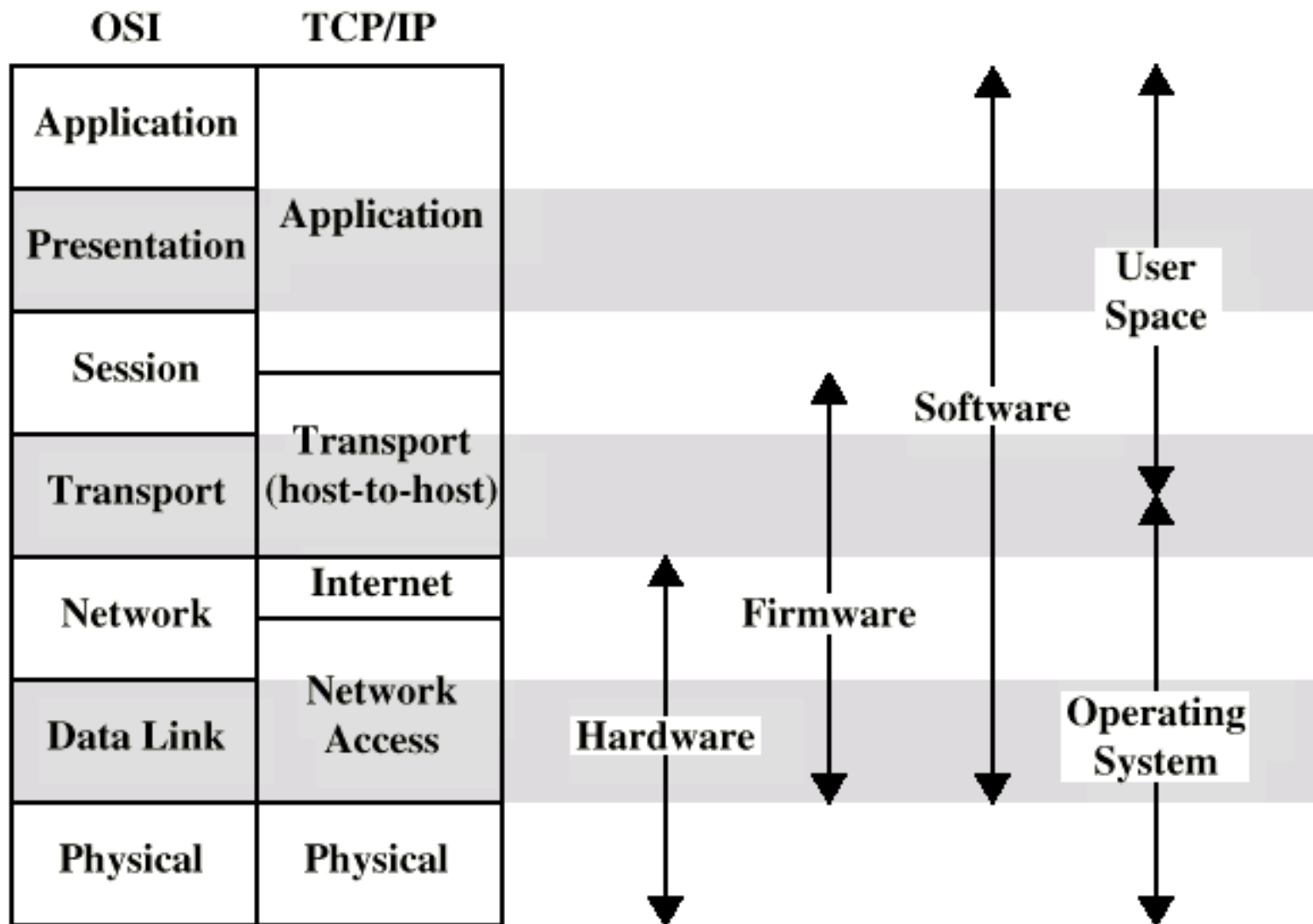
TCP/IP Protocol Architecture Model



PDU in TCP/IP



OSI v TCP/IP



Standards

- Required to allow for interoperability between equipment
- Advantages
 - Ensures a large market for equipment and software
 - Allows products from different vendors to communicate
- Disadvantages
 - Freeze technology
 - May be multiple standards for the same thing